

Generation of a construction planning from a 3D CAD model

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Abstract

In construction planning practice, increasingly 4D CAD system are applied for construction analysis and communication. Normally the planning expert is responsible for relating building components to construction activities. In this article we describe a method for automated generation of the construction planning. An algorithm is presented that derives the construction order from a solid model of the building. Experiences from a pilot study illustrate the differences between the real planning and the generated planning, and they show the limitations of the current implementation. Finally an outlook is presented on a more advanced planning system that includes contractor's specific knowledge for more accurate results.

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Keywords: 4D CAD; Construction management; Construction analysis; Automatic planning

1. Introduction

Traditionally, construction planning is a critical factor in building management. The construction planner, employed by a building contractor is a person with much experience in building construction who knows how to estimate the required labour and equipment from a building design. Using this knowledge a construction planning is created as the leading schedule for other derived plans such as transport, measurement, safety, etc. Project plans are constructed completely manually or using a specific tool like MS project[®] or Primavera[®].

Due to the critical factor of the planning, many research efforts have been directed to simulation of the building process based on the planning, to visually or computationally search for conflicts or errors [1,2]. From this research, 4D CAD systems have emerged, like InVizn[®], Navisworks[®] and 4D Suite[®]. These systems support the planner by relating building components from a 3D CAD system to construction activities from a project planning system, using a graphical interface. The construction process can then be simulated by executing the planning and the user can visually check how the process proceeds. 4D CAD systems can be used for construction analysis and communication. Experiences in practice by

Clayton [3] and Broekmaat [4] have learned that 4D CAD reduces the number of activities on the critical path, distributes the equipment more evenly, allows for more flexibility in time and reduces planning mistakes.

Even when using 4D CAD systems, the planning expert still plays a crucial role, as he manually relates building components to construction activities and visually tests whether problems occur during the construction process. 4D CAD systems do not bear any knowledge about the construction process itself. More advanced systems support collision detection, but there is no mechanism for expressing constraints and dependencies, and to test whether they are violated in the construction process.

In our research project we took up the challenge to automate the planning process, being well aware that a completely automated procedure is probably not feasible. However, this proposition allows us to investigate which construction knowledge can be derived from common sense or physical laws, which knowledge is construction method dependent, which knowledge is building component dependent and which knowledge cannot be described by the previous methods and thus remains for further investigation or in the hands of the planning expert. For knowledge representation we initially follow a computational approach to research how far this method can bring us in producing a reliable planning. A limited pilot study is executed to test the results against current planning practice. In this paper we report on construction

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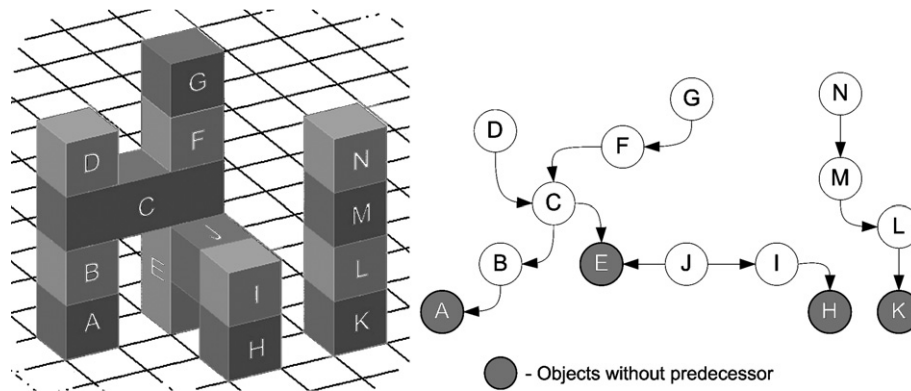


Fig. 1. 3D grid and graph schema.

algorithms we investigated and how well they performed. Finally we will discuss which deviations were found and how we think we can extend our system to produce a more reliable planning.

2. Construction algorithms

A construction algorithm encapsulates construction knowledge. Construction knowledge can be divided in knowledge about structural behaviour and knowledge about construction methods. Structural behaviour is determined by physical laws, and construction methods are determined by the building products and the construction equipment. For now, we assume that the construction method is fixed, which does not mean that our automated planning can handle only one construction method. We will come back to this issue in the discussion section. Given the construction method, there are still many construction orders possible. In this section we will discuss which computation methods are available for calculating the construction order of building components and how these methods perform.

An excellent overview of ordering methods is presented by Johnson et al. [5]. They differentiate between ordering by element type, by elevation, by joint configuration, and by tracing from foundation. The authors were not able to identify an algorithm that worked in a consistent, generalized manner. The methods flawed for special cases. They conclude that only Finite Element Analyses can solve this problem. In our project we take a different approach. We do not aim at an abstraction for structural analysis, but for construction analysis. Therefore we have researched algorithms that analyze the topology of a

building construction. Before that we will briefly discuss the basic principles for 3D geometry representation of building components, since this representation strongly affects the algorithmic approaches.

2.1. 3D representations

In building practice line drawings are still widely used. Recognition of building components from line drawings is in its infancy and is not considered in this research. Increasingly, architectural and engineering offices produce 3D models using surface modelling and/or solid modelling. With solid modelling, structural properties such as inertia and volume are readily available. CAD systems typically contain a library of building components that can be adjusted to the designer's needs. Building components (e.g. wall) can provide additional (e.g. material) and derived data (e.g. surface area).

2.2. Construction order analysis

Instead of calculating the vertical load distribution to find out which component bears another component, it is also possible to use the geometric data of the model. More specifically the topology of the building can tell which component is on top of or next to another component. Under the assumption that a 3D CAD model is available, we investigated two approaches for construction order analysis, namely 3D box representation and 3D solid representation. The analysis results in a directed graph, stored in a log file. In this graph the nodes represent the building components and from each node the edges point at components that are underneath or besides.

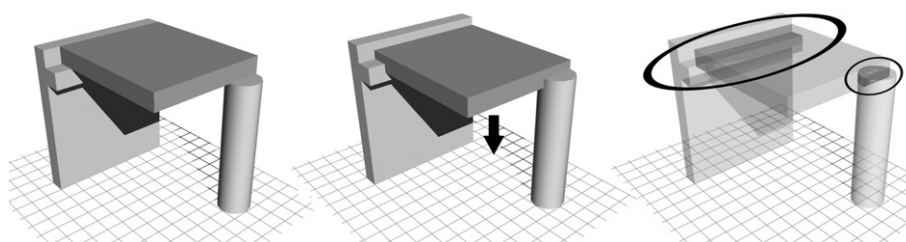


Fig. 2. 3D solid displacement and intersection.

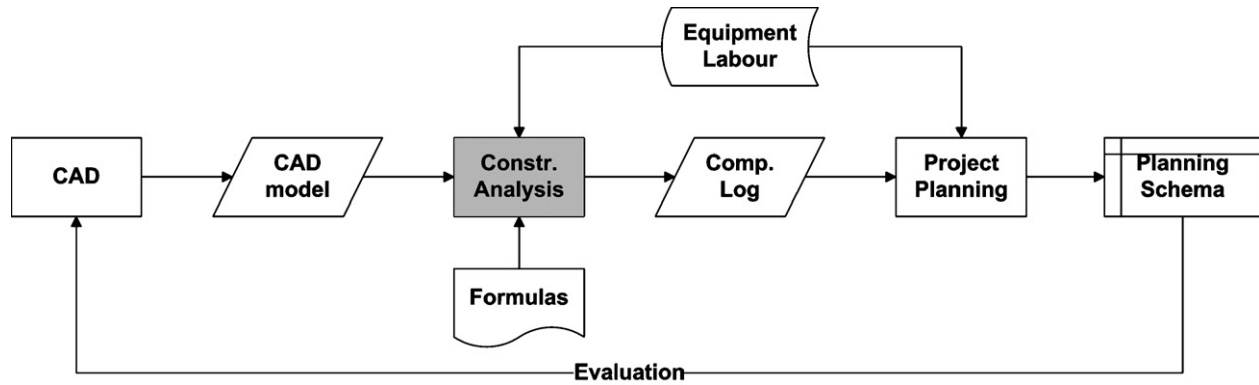


Fig. 3. Process model.

2.3. Adjacency inference

In our first approach we subdivided the original 3D model into a 3D grid. Building components represented, as boxes, can occupy one or more grid cells. The algorithm will test each component if any of the grid cells is vertically adjacent in one direction to other grid cells of other components (Fig. 1). If so, these vertical relationships will be registered in the log file. Following, for those components whose grid cells don't have any vertical adjacent neighbours, it will search for horizontal adjacent components in all four directions. These horizontal relationships are also registered in the log file. A useful grid size appeared to be 20 cm for obtaining good results with the construction order algorithm. The right grid size depends on the shape detailing of the building components. Most building components can be approximated with the 20-cm grid without losing essential details.

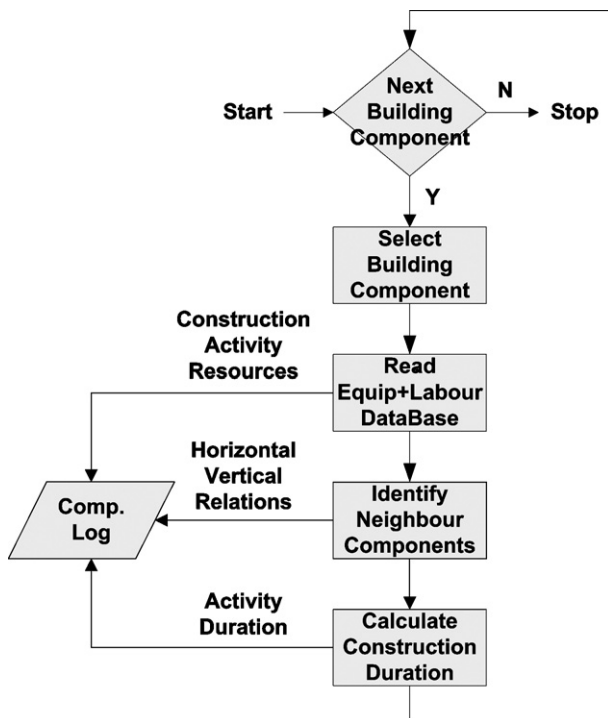


Fig. 4. Component Log generation process.

However, sometimes a finer grid is needed in case of very slim components such as steel frames.

Although this is a very elegant method that can process any 3D CAD model, it causes serious problems in terms of memory management. The minimal data needed are the cell centre coordinates for the occupied grid cells. Even a computer with a big RAM will fall short due to the memory demand of large amount of grid cells and relations that have to be traversed for an average building model to determine the building component topology. External storage could solve this problem, but will slow down the computation process dramatically.

Because of these problems we dropped the 3D grid approach and converted the original 3D model into a solid model. To determine the relationships between components, we took the following approach. First, each solid component is displaced vertically over a short distance (10 cm.) in the negative direction (i.e. downwards). After the displacement the Constructive Solid Geometry (CSG) intersection operation is used to find out which other components are intersected. Secondly, those components that did not vertically intersect any other component are displaced horizontally in all four directions. Again, the intersection operation is executed to find out what the neighbouring components are. From these results, similar to the 3D grid, the vertical and horizontal adjacency relationships are stored in a file for each component (Fig. 2).

This approach requires significantly less storage capacity and computing power, but is only applicable when a solid

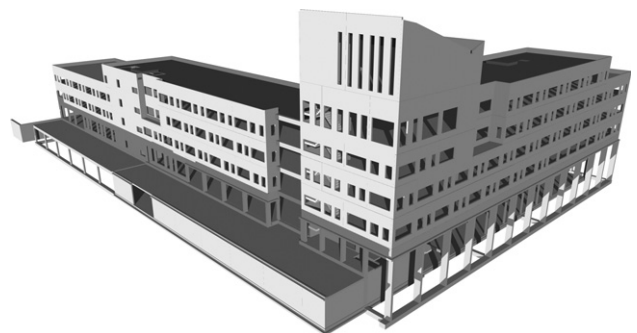


Fig. 5. Building for pilot study.

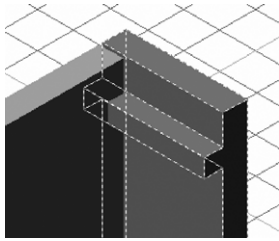


Fig. 6. Clash example.

modelling programme is used or conversion to a solid model is supported by the CAD program.

3. Implementation

In the proposed design and planning process, the construction analysis is preceded by the creation of a 3D model using a CAD program (e.g. Autocad/ADT). The construction analysis program imports the 3D model and derives the vertical and horizontal relationship between the building components as described in the previous section. A building component can occur in three situations: (i) only vertical relationships (e.g. walls, floors), (ii) only horizontal relationships (e.g. balconies) and (iii) vertical and horizontal relationship (e.g. stairs). To reduce the number of displacements during the construction order analysis, each component type is labelled with a displacement-direction attribute, indicating which displacement is appropriate. Identification of the construction component type is established in the current implementation by placing a component on a CAD layer with the right component type name. The construction analysis module reads from an external database of available equipment and labour, and formulas for the duration calculation. Reference data about construction time as a function of component length or volume, applied equipment and labour capacity can be found in literature [6]. During the construction analysis a component log is generated as presented in the next section. In the log file we find for each building component, its name, the duration of the construction and a list of components that it is horizontally or vertically related to. A project planning program (e.g. MS project) is used to import the log file and create a planning schema from it, i.e. all components will be listed subsequently and the appropriate resources (equipment and labour) will be allocated. The project planning program is also used to create PERT diagrams and for the calculation of the critical path.

3.1. Construction analysis module

The construction analysis module (see Fig. 3) is implemented in Autocad-Visual Basic for Applications (VBA) and proceeds as presented in Fig. 4.

From the Autocad-Architectural DeskTop (ADT) components that were used to model the building, the next item is selected. For this component type the equipment and labour data are read that are stored in the current implementation as a single table in an external XML file. The retrieved data are written as a new construction activity entry to the component relation and duration log. Next, using the 3D solid representation method as described in Section 2.3, the neighbours of the current component are identified for the displacement directions that are specified for this component type. The results are added to the log file as horizontal and vertical relationships. A boolean variable named workspace specifies for each component type whether workspaces around the building component are required for temporary constructions (e.g. scaffolds). If so, another building component must be in place (e.g. a floor) before the component under consideration can be constructed. In case the workspace attribute is set to true, then also the neighbouring components next to the building component directly below, are listed. In the last step the duration for the construction activity is determined by multiplying the calculated length or volume of the component with the reference data for equipment and labour capacity. This procedure is repeated until all components have been processed.

4. Pilot study

The pilot study (Fig. 5) is taken from practice and concerns the construction of an office building for the municipality in Amsterdam. From this project the 3D CAD drawings and the construction planning were available. The construction is a combination of prefabricated components and in-situ construction components.

4.1. 3D CAD model preparation

Despite the availability of the Autocad-ADT drawings, the CAD model was not ready for use in our construction analysis system. Since component type classification for construction analysis is absent, all ADT components (i.e. Autocad-objects) are transferred to a layer with the appropriate component type name (i.e. Autocad-layer), which is a manual procedure in the current implementation. Two additional computer programs

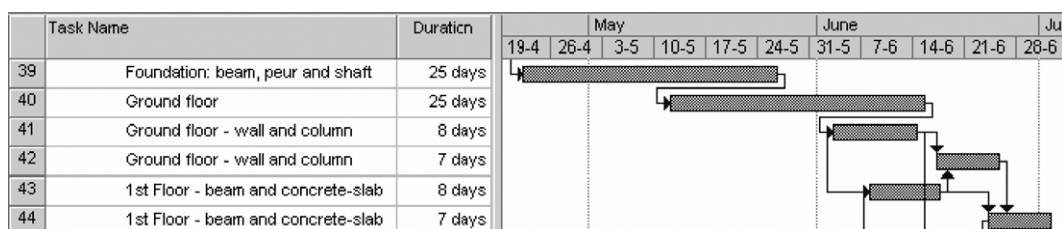


Fig. 7. Part of the real planning.

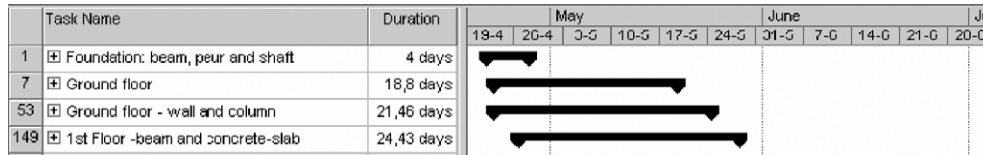


Fig. 8. Part of the generated planning.

were developed for model preparation. The first program converts the ADT components (e.g. wall, floor, etc.) into solids. Additional entities, such as annotations that are drawn using functions outside the ADT library are lost. The second program detects clashes between 3D solid components (Fig. 6). Inaccurate drawing of building components will produce incorrect results of the construction analysis program. The clash detection program will generate a list of building components that intersect and thus must be adjusted to prevent this error. In total the pilot study required 1 day preparation time.

4.2. Planning generation

After preparation of the CAD model, the generation of the planning is as straightforward as described in Section 3. For the case study the equipment and labour database is prepared in XML, listing for each component type the required resources per length or volume unit. In this test case these data were fixed for each building component. Consequently the system cannot search for construction alternatives. We will come back on this issue in the discussion section. Execution of the Autocad-VBA construction analyses script takes approximately 1 h for the test case on a standard PC.

4.3. Planning comparison

For obvious reasons we present here only a small part of the whole construction process. A comparison is made between the real planning (Fig. 7) and the generated planning (Fig. 8) of the concrete construction of the ground floor level.

First of all we notice that in the generated planning, the construction activities for all individual building components are grouped into bigger tasks. A group is defined as a collection of building components that are constructed subsequently and that share the same resources (equipment and labour). The original, real planning uses a slightly different grouping strategy, based upon experience, to identify the bigger tasks without detailing into subtasks.

There is a striking difference between the generated duration of the foundation task and the really planned duration. This difference can be explained by two factors: (i) in the generated planning all foundation construction activities can be executed concurrently because of unlimited resources, and (ii) the contractor applied a different foundation technique than the one selected from the resources database. These two factors introduce a major difference between the overall generated planning and the real planning for the complete project. Because the generated planning is created under the assumption

of unlimited resources, it will be the fastest construction possible by definition. Only through the selection of another construction type with other data on required resources per unit, the overall construction time can be shortened.

Another interesting result is that in the generated planning, the construction of the walls and columns of the ground floor starts at the same time as the construction of the ground floor slab. The reason is that the floor slabs are supported by the foundation within the walls and thus a wall component can be put in place before the floor is constructed. Walls and columns construction are completely independent from the floor construction (Fig. 9).

Finally we conclude that the construction order is correct. First the foundation is constructed, then the ground floor and the walls and columns. Shortly after the construction of the ground floor walls and columns, the construction of the first floor can start. However, the overall construction time was significantly shorter because of the availability of unlimited resources in the generated planning.

5. Discussion

The current system lacks construction knowledge. The construction analysis program will execute a construction activity as soon as this is feasible. There is no knowledge of equipment and labour capacity, costs, material storage capacity and location, preferences of the contractor, etc. However, we anticipate that much of this additional contractor specific knowledge can be expressed as constraints in XML. Obviously the construction analysis component needs to be adapted for interpretation of more complex and dynamic constraints (e.g. crane movement).

The outcome of the generated planning sometimes raises questions about the traditional construction method, such as the case of the ground floor slab in the previous section. This result can be a planning flaw because of lack of construction

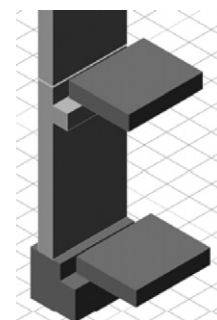


Fig. 9. Slab position.

knowledge (e.g. the floor slabs are required for the craftsmen to position the wall and columns) but it can also reveal new options for concurrent building that we have been unaware of (e.g. wall and column positioning by construction cranes or robots).

Confronted with our prototype system, the project planners responded quite positively. Although the results were not correct they judged the tool useful for a first quick estimate of the planning especially in the tendering stage where information is often still incomplete. To get more control over the application they expressed a need for a layered approach that allows for planning on different abstraction levels dependent on the stage of the design process.

Most importantly, the proposed system supports the evaluation of alternative construction methods. After a first analysis using the building components layout as designed and default construction capacity values for equipment and labour, the resulting planning schema can be investigated. In the following evaluation runs all variables (building component type, equipment, labour) can be varied to study the effect in time, and required resources (equipment capacity, labour capacity). This is represented by the evaluation feed back loop in Fig. 3. Given a number of different construction methods, even one step further this evaluation process can be extended using optimisation methods from operations research (e.g.

linear programming and genetic algorithms). In this way the system can become a construction analysis tool not only for construction planning experts, but for designers as well, who want to study the various effects of constructive solutions as part of the design process.

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